

ARPAN CHATTERJEE

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EDUCATION

M.S in Computer Science, New York University

[New York, NY] [Sept 2022 - May 2024]

Relevant Coursework: Deep Learning, Data Structures and Algorithms, Cloud Computing, Big Data, Computer Networks

B. Tech in Computer Science and Engineering, NIT Durgapur

[Durgapur, India] [July 2016 - May 2020]

SKILLS

- **Languages:** C++, Java, Python, Javascript, PHP, Shell
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Web:** HTML, CSS, Django, Flask, React, Node, REST API, GraphQL, TCP/IP, HTTP/HTTPS, gRPC, WebRTC
- **Cloud:** AWS, Azure, GCP, Availability, Scalability, Reliability, Distributed Systems
- **Data:** Hadoop, Spark, Kafka, RabbitMQ, ETL, ElasticSearch, Airflow, SQL, NoSQL, Warehouse, Flink
- **AI/ML:** PyTorch, CUDA, RAG, LLM, NN, Fine-tuning
- **DevOps/SRE:** Jenkins, Ansible, Git, GitHub Actions, Terraform, Docker, Kubernetes, Perforce, CI/CD, Automation, Monitoring, Logging, Alerting, Grafana
- **AWS:** EC2, S3, RDS, DynamoDB, IAM, SQS, ECS, API Gateway, Kinesis, SageMaker, CloudFormation, Load Balancer, Auto Scaling, Cognito
- **Software Development Tools:** Confluence, Jira, VIM, Emacs, VisualStudio, Notion, Slack

PROFESSIONAL EXPERIENCE

NYU Vital Labs, *Software Engineer Assistantship*

[Feb 2023 - Sept 2023]

- Deployed a cloud infrastructure of virtual machines for over 100 students using Nginx, Django, React and Postgres.
- Integrated Redis to streamline mapping of VM tokens to Xen hypervisor port numbers for VNC client operations, and implemented RabbitMQ for asynchronous VM status updates, cutting VM retrieval time by over 35%.
- Utilized InfluxDB to monitor key metrics from web servers and hypervisors, including CPU utilization and memory usage, and integrated this data into a Grafana GUI, enhancing system administration for the infrastructure.

Qualcomm, *Software Engineer*

[July 2020 - July 2022]

- Designed, developed, and maintained the large-scale infrastructure of CI/CD integration, compilation and release pipelines of multiple modem FW modules in Jenkins using Python and Bash, following Agile and SDLC best practices.
- Compiled various tools and scripts into a unified, reusable binary, forming a Python package to streamline the deployment process. Migrated the entire infrastructure to Kubernetes as services.
- Developed a robust tool using Splunk API to retrieve logs and generate insights, improving team efficiency and aiding over 100 developers in code debugging.
- Devised an automated testing application development in Python to select an optimal set of test cases based on test coverage by functions, reducing test time by 90%, and leading to its adoption across departments.
- Spearheaded the development and execution of thousands of test logs in AWS ES. Deployed Kibana to visualise and analyze logs by employing Elasticsearch query DSL for sophisticated analysis.

IIT Kharagpur, *Research Intern*

[May 2019 - May 2020]

- Conducted concurrent data collection and processing using multiple threads from 280,000 Linux bug pages for information retrieval and data mining to recommend developers with pages to resolve the bugs.

PROJECTS

- Developed a robust search engine using a compressed variable-byte-encoded inverted index in C++ on 3 million web pages for quick query processing. Computed BERT embeddings to retrieve a query's top 10 recommended pages.
- Engineered a distributed low latency book recommendation system app (having user authentication) integrating 13 AWS services based on user preference and activities. Enhanced scalability, fault tolerance, and seamless orchestration by creating a Docker container for the Django app and deploying it on Kubernetes (AWS EKS).
- Researched methods in optimising node localisation using deep neural networks utilizing the TDOA approach, with findings published at Springer.